



USN

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

RI2002-1 – DESIGN OF ROBOTIC COMPONENTS

(For RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: (i) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

(ii) **Use of Design Data hand Book is permitted.**

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **Where in the stress-strain curve, the hooke's law is valid?**
A) Strain hardening region B) Necking region
C) Elastic range D) Valid everywhere
2. **Which of the following factors are not considered while selecting values for factor of safety?**
A) Cost B) Failure consequences
C) Degree of uncertainty about loading and material D) None of the above
3. **The process of creating mechanisms and shapes of mechanical elements for a machine to get the desired output for a given input is called as**
A) Analysis B) Innovation
C) Synthesis D) Designing
4. **Use of multiple notches in a V shaped flat plate will**
A) Reduce the stress concentration B) Increase the stress concentration
C) No effect D) Cannot be determined
5. **Where is the necking region?**
A) The area between lower yield point and upper yield point B) The area between the plastic limit and elastic limit
C) The area between the ultimate point and initial point D) The area between the ultimate point and rupture
6. **Modulus of resilience is defined as**
A) Strain energy per unit volume B) Strain energy per unit area
C) Independent of strain energy D) None of the mentioned
7. **In design process, which step is followed after defining the problem?**
A) Analysis B) Synthesis
C) Optimization D) Evaluation
8. **Which of the following reduces the stress concentration?**
A) Use of multiple notches B) Drilling additional holes
C) Removal of undesired material D) Each of the mentioned
9. **If a fastener is threaded into a tapped hole, then the fastener is likely to be called as**
A) Screw B) Bolt
C) Washer D) Screw or bolt
10. **If there is no place to accommodate the nut, then one would choose the**
A) Through Bolts B) Tap Bolts
C) Studs D) None of the mentioned

11. The power transmitted by a belt is maximum when the maximum tension in the belt (T) is equal to
 - A) T_c
 - B) $2T_c$
 - C) $3T_c$
 - D) $4T_c$
12. The centrifugal tension in belts
 - A) Increases power transmitted
 - B) Decreases power transmitted
 - C) Have no effect on the power transmitted
 - D) Increases power transmitted upto a certain speed and then decreases
13. When the belt is stationary, it is subjected to some tension, known as initial tension. The value of this tension is equal to the
 - A) Tension in the tight side of the belt
 - B) Tension in the slack side of the belt
 - C) Sum of the tensions in the tight side and slack side of the belt
 - D) Average tension of the tight side and slack side of the belt
14. Which of the following cannot take radial load?
 - A) Cylindrical Roller bearing
 - B) Taper roller bearing
 - C) Thrust ball bearing
 - D) None of the listed
15. In angular contact bearings, ____ bearings are required to take thrust load in both directions.
 - A) 1
 - B) 4
 - C) 2
 - D) 3
16. The angular play bearing must be mounted without axial play.
 - A) Yes
 - B) Little tolerance is adjusted
 - C) Little tolerance is necessary
 - D) Can't be stated
17. Flexible shafts have _____ rigidity in torsion making them flexible.
 - A) Low
 - B) High
 - C) Very high
 - D) Infinitely small
18. Is it necessary for an axle to be _____ with respect to rotating element?
 - A) Stationary
 - B) Moving
 - C) Moving or stationary
 - D) None of the listed
19. Among the normal module and transverse module, which one has greater value?
 - A) Normal Module
 - B) Transverse Module
 - C) Both have equal module
 - D) Insufficient information
20. Which of the following can be used for power transmission in intersecting shafts.
 - A) Spur Gear
 - B) Helical Gear
 - C) Bevel Gear
 - D) None of the listed

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Unit – I		Marks	BT*	CO*	PO*
1.	a) With salient features explain the Stress-strain diagram for brittle and ductile materials.	8	L*1	1	1
	b)* Determine the maximum normal stress and maximum shear stress at section A-A for crank shown in Fig.Q1 (b). When load of 10kN is assumed to be concentrated at the center of crank pin.				

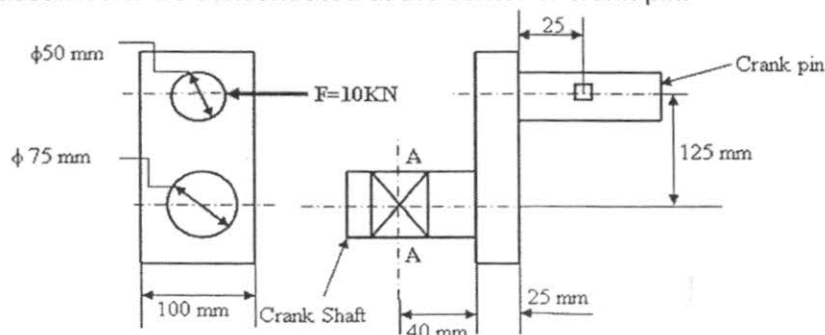


Fig.Q1 (b)

8 L2 2 1

2. a) A metallic bar 250mmX80mmX30mm is subjected to a force of 20 kN (Tensile), 30kN (Tensile), and 15 kN (Tensile) along the x, y, and z directions respectively as shown in Fig Q 2 (a). Determine the change in the volume of the block. Take $E=2 \times 10^5 \text{ N/mm}^2$ and poisons ratio = 0.25

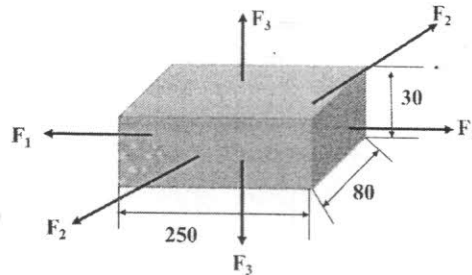


Fig Q 2 (a).

- b) A plate of 45C8 steel ($\sigma_{yt}=353\text{MPa}$) is subjected to the following stresses. $\sigma_x=150\text{N/mm}^2$, $\sigma_y=100\text{N/mm}^2$, $\tau_{xy}=50\text{N/mm}^2$. Find the factor of safety by
- Maximum Principal Stress Theory
 - Maximum Shear Stress Theory
3. a) A hollow steel tube will carry an axial compressive load of 150 kN. The yield stress for steel is 250MPa and a factor of safety of 2. The following 3 classes of tubes of an external diameter of 100 mm are available.

Sl No	Class	Thickness
1.	Light	3.5mm
2.	Medium	4mm
3.	Heavy	5mm

Suggest Suitable Section

- b) A flat bar as shown in Fig. Q 3 (b) carries an axial load of $P=25\text{KN}$ determine the maximum stress in the member. If it is made up of C40 steel, what is the value of FOS? If it C30 steel and desired FOS is 2.5. What is the value of thickness t for the bar.

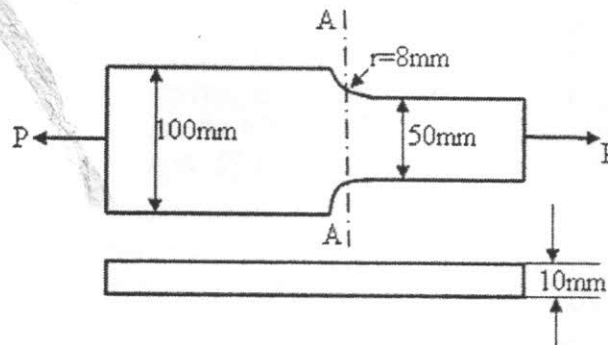


Fig. Q 3 (b)

Unit – II

4. a) A cylinder of effective diameter 250mm is subjected to a pulsating pressure that varies from zero to a maximum of 1.5MPa. The cylinder head is connected by means of 8 bolts. The endurance limit of the bolt material is 208MPa and the yield strength is 330MPa. The bolts are tightened with an initial preload of 1.5 times that of the maximum steam load. A soft copper gasket is used to make the joint leakproof. Assume a stress concentration factor of 2.8 and a factor of safety based on yield as 2. Find the required size of bolts.

- b) List the merits and Demerits of flat belt drive.

8 L2 4 1

5. a) A pulley bracket as shown in Fig Q 5 (a) is supported by 4 bolts, two at A-A and two at B-B. Determine the size of bolts using an allowable shear stress of 25 N/mm² for the material of the bolts.

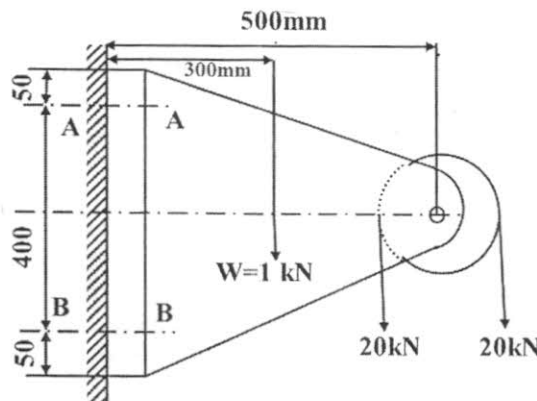


Fig Q 5 (a)

- b) What are the advantages and disadvantages of roller bearings?

8 L2 3 1
8 L2 4 1

6. a) Select a wire rope to lift a load of 10 kN through a height of 600 m from a mine. The weight of the bucket is 2.5 kN. The load should attain a maximum speed of 50 m/min in 2 seconds. Take $F_u = 500.8d^2$.
b) Select a suitable single-row radial ball bearing to carry a radial load of 1.5 kN and a thrust load of 1.2 kN at 900 rpm. The bearing is to be used 7 hours per day and an average service life of 8 years is desired. Consider the design load for bearing during selection with speed factor, life factor, trust factor, and application factor.

8 L2 3 1
8 L2 4 1

Unit - III

7. A steel solid shaft transmits 15 kW at 200 rpm is supported on two bearings 750 mm apart and has two gears keyed to it. The pinion having the 30 teeth of a 5 mm module is located 100 mm to the left of the right-hand bearing and delivers power horizontally to the right. The gear having 100 teeth of a 5 mm module is located 150 mm to the right of the left-hand bearing and receives power in a vertical direction from below. Using allowable stress of 54 MPa in a shear, determine the diameter of the shaft. Take $C_m = 1.75$, $C_t = 1.25$ and $\alpha = 18^\circ$.
8. a) Sketch and explain the different terminology of spur gear.
b) A pair of spur gears consists of a 20 teeth pinion meshing with a 120 teeth gear. The module is 4 mm. Calculate (i) The centre distance (ii) the pitch circle diameters of the pinion and the gear (iii) The addendum and dedendum (iv) The tooth thickness.

16 L3 5 1
8 L2 5 1
8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
